FEEDBACK / PID

A quick PI stability check

With unity negative feedback, use G(s)=1/(s+1) and C(s)=1+Ki/s, where Ki>0.

Is the closed loop asymptotically stable for every positive Ki? Justify briefly.

TW-PID-01 Graduate control

Suggested: 1-2 min. Justify your answer.

FEEDBACK / PID

Why does the integrator wind up?

An actuator saturates while a PI controller keeps accumulating positive error.

Why can recovery become slow, and how can anti-windup help?

TW-PID-02 Graduate control

Suggested: 1-2 min. Justify your answer.

STABILITY

A Lyapunov derivative with zeros

For dx/dt=y and dy/dt=-x-y^3, V=(x^2+y^2)/2 gives dV/dt=-y^4.

Do all trajectories approach the origin? Explain why y=0 cannot persist when x is nonzero.

TW-STA-01 Graduate control

Suggested: 1-2 min. Justify your answer.

STABILITY

Which initial states return to zero?

For dx/dt=-x+x^3, which initial states converge to zero: |x(0)|<1, |x(0)|<=1, or all real values?

Explain your choice.

TW-STA-02 Graduate control

Suggested: 1-2 min. Justify your answer.

LINEAR SYSTEMS

An uncontrollable but stabilizable system

A continuous-time system has A=diag(-1,2), B=[0;1], and feedback u=-Kx.

Can feedback stabilize it even though it is not controllable? Give one suitable K.

TW-LIN-01 Graduate control

Suggested: 1-2 min. Justify your answer.

LINEAR SYSTEMS

Can one input control both states?

A continuous-time system has A=[0,1;0,0], B=[0;1].

Is it controllable? Show the matrix [B,AB].

TW-LIN-02 Graduate control

Suggested: 1-2 min. Justify your answer.

ESTIMATION

How much should a Kalman update move?

For y=x+v, a scalar Kalman filter has prior estimate x_hat=2, prior variance P=1, and independent measurement-noise variance R=3. The new measurement is y=6.

What is the updated estimate?

TW-EST-01 Graduate control

Suggested: 1-2 min. Justify your answer.

ESTIMATION

Can state and bias be separated?

A sensor repeatedly reports y=x+b, where both the unknown state x and unknown bias b are constant.

Can an estimator uniquely recover both from these readings alone? Explain.

TW-EST-02 Graduate control

Suggested: 1-2 min. Justify your answer.

OPTIMIZATION / MPC

A bounded control move

For x_next=x+u at x=2, minimize (x+u)^2+u^2 subject to -0.5<=u<=0.5.

What input minimizes the cost?

TW-MPC-01 Graduate control

Suggested: 1-2 min. Justify your answer.

OPTIMIZATION / MPC

What does receding horizon mean?

An MPC solve returns the planned inputs [0.4, 0.1, -0.2].

In standard receding-horizon control, which input is applied now, and what happens at the next sampling instant?

TW-MPC-02 Graduate control

Suggested: 1-2 min. Justify your answer.

MULTI-AGENT SYSTEMS

Connected, but still no consensus

Two agents synchronously use x1(k+1)=x2(k), x2(k+1)=x1(k). Initially their values are 0 and 2.

Do they converge to consensus? Explain.

TW-MAS-01 Graduate control

Suggested: 1-2 min. Justify your answer.

MULTI-AGENT SYSTEMS

Consensus can be a weighted average

Agents use x(k+1)=W*x(k), with W=[1/2,1/2;1/4,3/4], x(0)=[0;6].

What common value do they approach? Briefly justify.

TW-MAS-02 Graduate control

Suggested: 1-2 min. Justify your answer.